



**BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE
APPLICATIONS AND GEO-INFORMATICS**

WEEKLY PROGRESS REPORT (18/05/2026 – 24/05/2026)

WEEK 1

PROJECT NAME

ROAD CONDITION DETECTION

PROJECT DESCRIPTION:

**AUTOMATIC DETECTION AND
SEGMENTATION OF ROAD DAMAGES**

GROUP MEMBERS:

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GROUP ID:

2026G263

GROUP GUIDE:

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GROUP COORDINATOR:

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DATE	
18/05/2026	- Completed internship registration and introductory session. - Discussed available project domains with mentors, including road damage detection and video analytics-based solutions
19/05/2026	- Researched model architectures and inference pipeline concepts. - Studied available datasets, annotation formats, and preprocessing methods.
20/05/2026	- Worked on Python environment setup and explored inference workflow. - Explored and prepared small dataset structure, labelling methods, and training configurations.
21/05/2026	- Initiated model training using the prepared dataset and configured training parameters. - Successfully tested the trained model on a sample video for road damage detection.
22/05/2026	- Explored object tracking concepts and integrated track ID functionality with the detection pipeline. - Refined video inference workflow and analysed tracking output for moving objects.
23/05/2026	Holiday (4th Saturday)
24/05/2026	Holiday (Sunday)

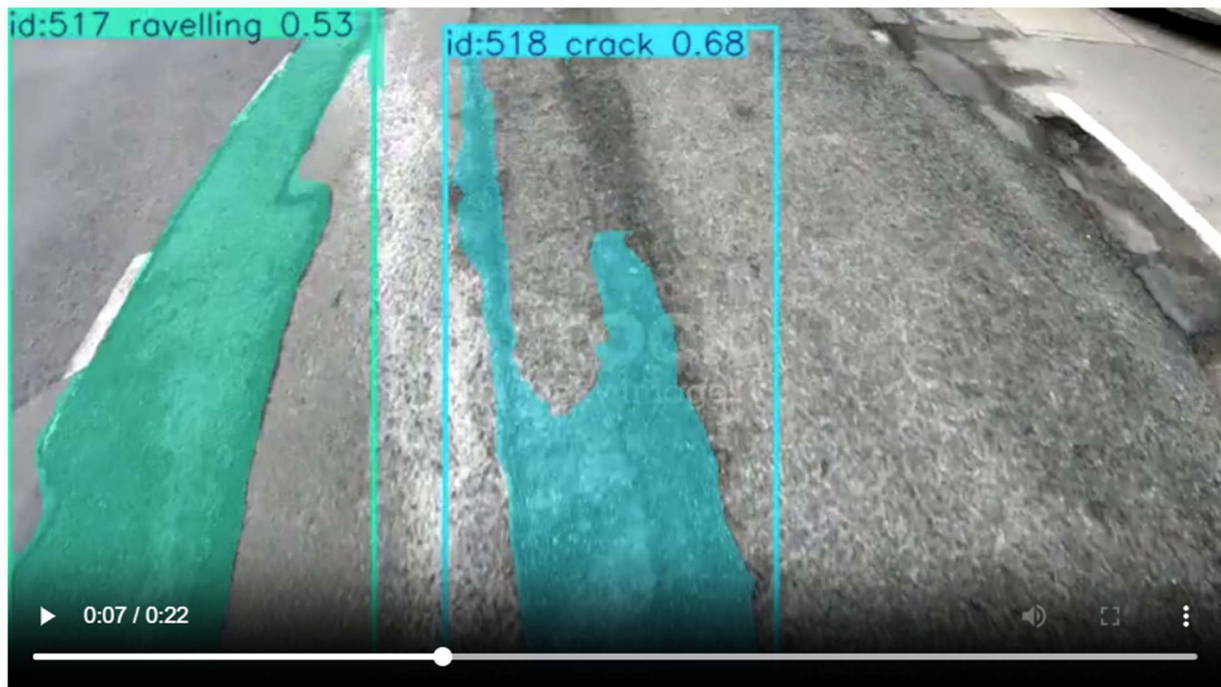
Next Week Plan	Next week, the focus will be on improving dataset annotations and refining the training process for better detection and tracking performance. We will continue testing the model on different real-world road videos, optimize detection for nearby objects, and further improve the tracking system to ensure accurate counting and monitoring of potholes and road damages without duplication.
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REFERENCE:

- [Ultralytics YOLO11 Documentation](#)
- [Ultralytics GitHub Repository](#)
- [ByteTrack GitHub Repository](#)
- [Roboflow Official Website](#)
- [Roboflow Pothole Dataset](#)
- [Pothole Segmentation Dataset](#)
- [YOLO11 Research Paper \(arXiv\)](#)
- [YOLO11 Training Tutorial \(YouTube\)](#)

SCREEN SHOTS:





Dataset [How to Search](#)

[Train Model](#)

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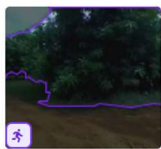
Classes

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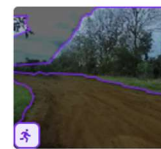
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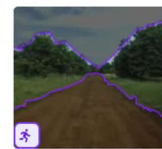
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